DO NOW — RECALL

# Paper 2 extended response challenge

Name: _____ Class: _____ Date: _____

**1. What is a precondition?**

**2. What is a loop invariant?**

**3. What does O(1) auxiliary space mean?**

**4. For [1,1,2,3,3], count globally distinct values and consecutive runs.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Requirement that must be satisfied for correctness.

### 2. Model answer

Property maintained as iterations progress.

### 3. Model answer

Additional memory remains constant with input size.

### 4. Model answer

Three distinct values; three consecutive runs.

### By the end, you can…

• State the sorted-input precondition and any edge cases.

• Trace changing values with a loop invariant and correct bounds.

• Explain why the selected algorithm is efficient and when another is preferable.

### Vocabulary — know these terms

precondition — Required assumption about input.

invariant — Property maintained by a loop.

auxiliary space — Memory used beyond input.

hash set — Structure for expected efficient membership tests.

NEW LEARNING

# Read, annotate and explain

### Step 1

State the sorted-input precondition and any edge cases.

### Step 2

Trace changing values with a loop invariant and correct bounds.

### Step 3

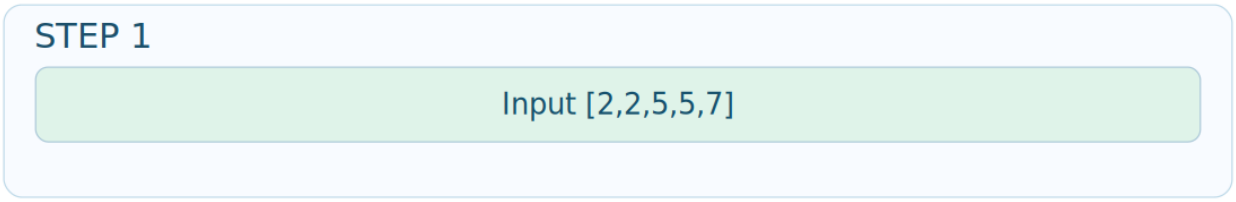
Explain why the selected algorithm is efficient and when another is preferable.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

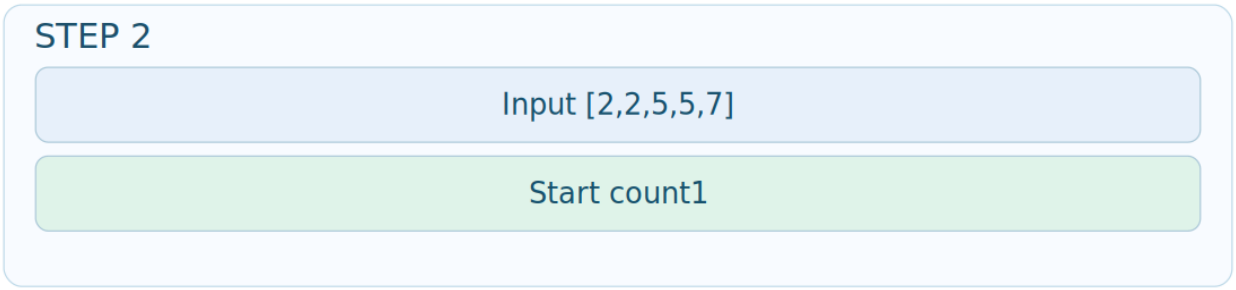
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**



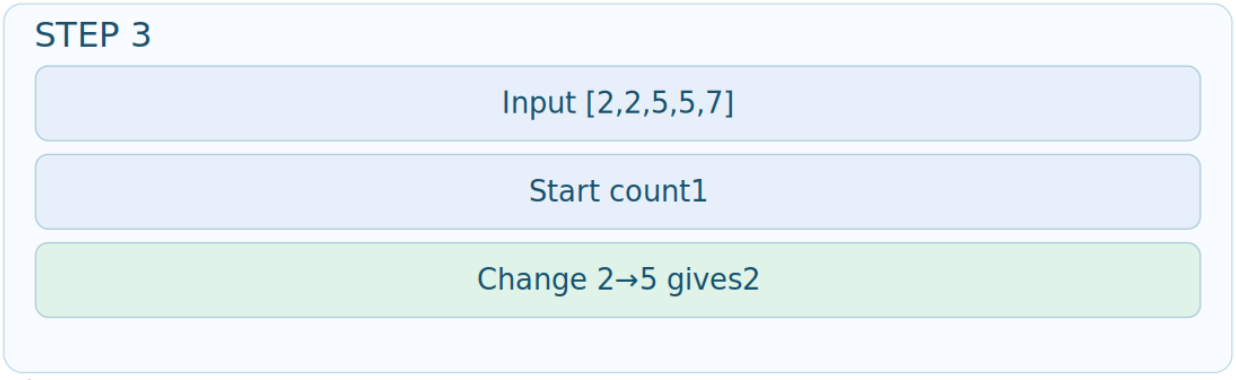

STEP 1

**New Learning**




STEP 2

**New Learning**

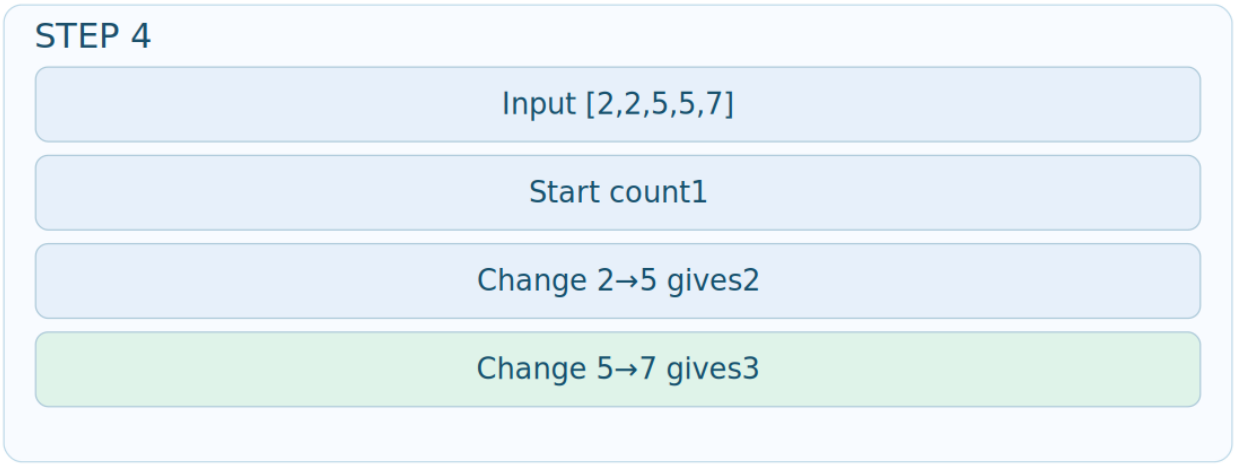



STEP 3

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




STEP 4

I DO — WORKED EXAMPLE

# Follow the worked method

### Trace UniqueCount for [1,1,2,3,3].

### Explain the worked method

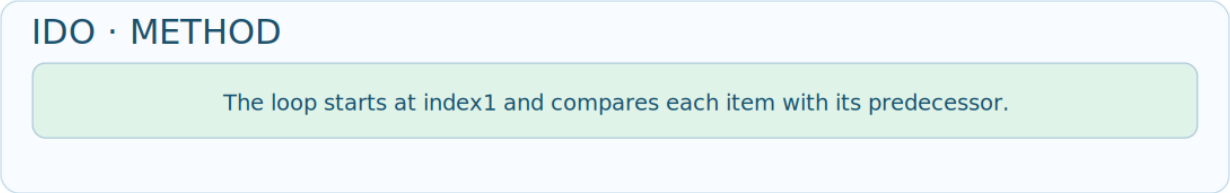
The loop starts at index1 and compares each item with its predecessor.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Trace UniqueCount for [1,1,2,3,3].

The loop starts at index1 and compares each item with its predecessor.




IDO · METHOD

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Trace UniqueCount for [4,4,4,9,10,10].
Source code / input:
Return3; groups 4,9,10.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Write pseudocode to count distinct values in a sorted array, including empty input. [8]

### Question 2

Trace [2,2,5,5,6,9,9] and give time/space complexities. [9]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Evaluate adjacent counting versus a hash set when the data is unsorted and contains a million elements. [9]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**An adjacent comparison algorithm for sorted input works identically on unsorted input.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

## GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

## Guided 1 — CHECK

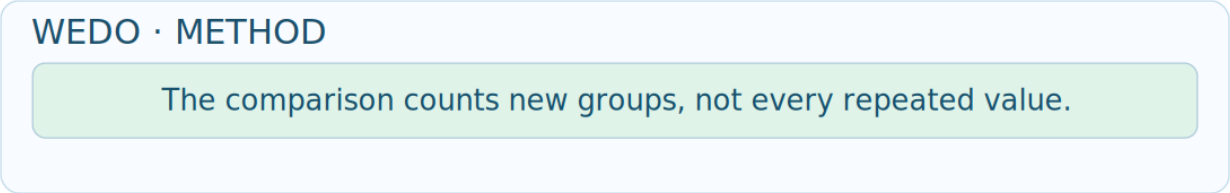
Explain the worked method
The comparison counts new groups, not every repeated value.

# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Trace UniqueCount for [4,4,4,9,10,10].**

The comparison counts new groups, not every repeated value.




WEDO · METHOD